# Week 7: More proofs w/ sets

$A \subseteq B$ means $x \in A \Rightarrow x \in B$ "Let $x \in A$. ..... Thus $x \in B$."

$\forall x \in A, x \in B$

$A \cap B = \{x : x \in A \text{ and } x \in B\}$

$A \cup B = \{x : x \in A \text{ or } x \in B\}$

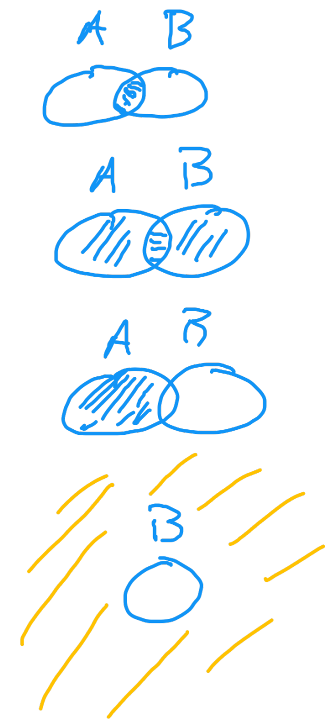
$A - B = \{x \in A \mid x \notin B\}$

$\overline{B} = \{x : x \notin B\}$

$=$

$U - B$

$U =$ "everything"



Prove or disprove:

(i) $6\mathbb{Z} = 2\mathbb{Z} \cup 3\mathbb{Z}$ F

(ii) $6\mathbb{Z} = 2\mathbb{Z} \cap 3\mathbb{Z}$ T

$d\mathbb{Z} = \{n \in \mathbb{Z} \text{ s.t. } d \mid n\}$

Proof:

(i) This is false: $2 \in 2\mathbb{Z} \cup 3\mathbb{Z}$, but $2 \notin 6\mathbb{Z}$.

(ii) "$=$" is "$\subseteq$" and "$\supseteq$"

"$\subseteq$" Let $x \in 6\mathbb{Z}$. Then $x \in \mathbb{Z}$ and $6 \mid x$.

(WTS: $x \in 2\mathbb{Z} \cap 3\mathbb{Z}$, i.e., $x \in 2\mathbb{Z}$ and $x \in 3\mathbb{Z}$, i.e., $x \in \mathbb{Z}$, $2 \mid x$ and $3 \mid x$)

Since $2 \mid 6$ and $3 \mid 6$, by transitivity of division, $2 \mid x$ and $3 \mid x$.
Thus $x \in 2\mathbb{Z}$ and $x \in 3\mathbb{Z}$, so $x \in 2\mathbb{Z} \cap 3\mathbb{Z}$.

"$\supseteq$" Let $x \in 2\mathbb{Z} \cap 3\mathbb{Z}$. Then $x \in 2\mathbb{Z}$ and $x \in 3\mathbb{Z}$. Then $x \in \mathbb{Z}$ and $2 \mid x$ and $3 \mid x$.

(WTS: $x \in 6\mathbb{Z}$, i.e., $6 \mid x$).

Since $\gcd(2,3) = 1$, $2 \cdot 3 \mid x$. Thus $x \in 6\mathbb{Z}$. ∎

$P \Rightarrow (Q \Rightarrow R)$

negation

$\neg\, ($

$x \in A - C \Rightarrow x \in A - B$

$B \subseteq C \wedge \exists x \in A - C$ s.t. $x \notin A - B$

$\qquad P \Rightarrow Q$

**Claim:** $(B \subseteq C) \Rightarrow (A - C \subseteq A - B)$

**Proof:** ~~Let $x \in B$.~~ ! #1 wrong answer

Assume $B \subseteq C$. Let $x \in A - C$. Then $x \in A$ and $x \notin C$.

(WTS: $x \in A - B$, i.e., $x \in A$ and $x \notin B$).

Proceed by contradiction. Assume $x \in B$. Since $B \subseteq C$, $x \in C$. This contradicts $x \notin C$. Thus $x \notin B$. Thus $x \in A - B$. ⊠

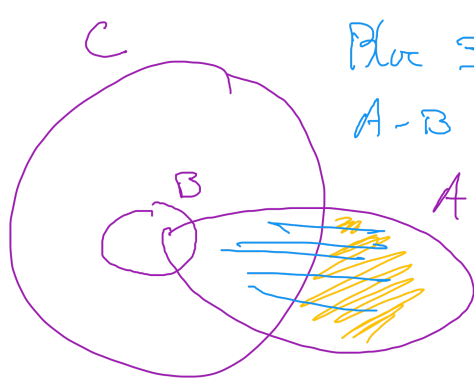
ALT. Proof.

... $x \notin C$. By the contrapositive of $B \subseteq C$, $x \notin B$.

$(x \in B \Rightarrow x \in C)$

Note: The contrapositive of "$B \subseteq C$" is

$x \notin C \Rightarrow x \notin B$




Blue ⊇ Yellow

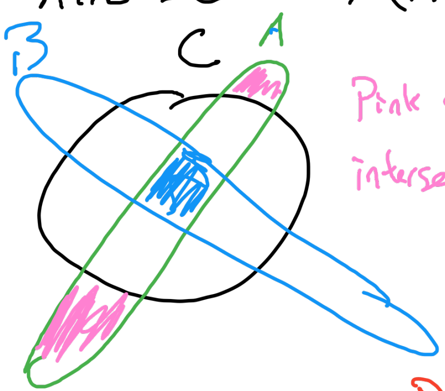
$A - B \supseteq A - C$

$P \Rightarrow Q$

**Claim:** $A \cap B \subseteq C \Rightarrow (A-C) \cap B = \phi$

To prove "$D = \phi$", do proof by contradiction.

"$D \neq \phi$", i.e. $\exists x \in D$.




Pink does not intersect B

Fine, but not ideal

"Let $x \in A - C \cap B$

..... $x \in \phi$"

**Proof:** $\boxed{\text{Let } \cancel{x \in A \cap B}}$

Don't do this.

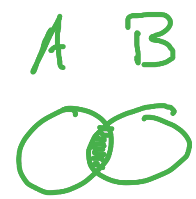
Suppose $A \cap B \subseteq C$. Proceed by contradiction. Assume $(A-C) \cap B \neq \phi$. Thus $\exists x \in (A-C) \cap B$. Then $x \in A - C$ and $x \in B$. Then $x \in A$ and $x \notin C$. Thus $x \in A \cap B$, so since $A \cap B \subseteq C$, $x \in C$. This is a contradiction, thus $A - C \cap B = \phi$. $\boxtimes$

# De Morgan's Laws

$\overline{C} = \{x : x \notin C\}$

(*) $\overline{A \cap B} = \overline{A} \cup \overline{B}$

$\overline{A \cup B} = \overline{A} \cap \overline{B}$





**Proof:** "$\subseteq$" Let $x \in \overline{A \cap B}$. Then $x \notin A \cap B$.

(WTS $x \in \overline{A} \cup \overline{B}$ i.e.

$x \in \overline{A}$ or $x \in \overline{B}$ i.e.

$x \notin A$ or $x \notin B$

$= \neg(x \in A \text{ and } x \in B) = \neg(x \in A \cap B)$)

Thus $x \notin A$ or $x \notin B$. Thus $x \in \overline{A}$ or $x \in \overline{B}$.

Thus $x \in \overline{A} \cup \overline{B}$. $\boxtimes$

`$ A \times B $`

## Products + Power Sets

Defn: let $A$ and $B$ be sets. The Cartesian Product $A \times B$ is the set $\{(a,b) : a \in A \text{ and } b \in B\}$.

Let $n \in \mathbb{N}$. then $A^n = \{(a_1, \ldots, a_n) \text{ s.t. } \forall i \in \{1, \ldots, n\}, a_i \in A\}$

$(A \times B \times C \ni (a,b,c), a \in A, b \in B, c \in C)$

Order matters $A \times B \neq B \times A$

$\boxed{\text{But}}$ $A$ and $B$ can be different

1. $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2 \ni (x,y), \; x,y \in \mathbb{R}$
2. $A = \{1,2\}, \; B = \{3,4\}$

$$A \times B = \left\{ \begin{matrix} (1,3), (1,4) \\ (2,3), (2,4) \end{matrix} \right\}$$

$(1,2) \notin A \times B$

$A \times B \neq B \times A$

$(1,3) \in A \times B, (1,3) \notin B \times A$ b/c $1 \notin B$

3. $(a,b) \in A \times B \iff a \in A \text{ and } b \in B$
4. $\{-1,1\} \times \mathbb{Z} \ni (1,7)$
5. $\mathbb{R} \times \mathrm{Fun}(\mathbb{R},\mathbb{R}) \ni (\pi, f)$

$1 \in \mathbb{Z}$ $\quad$ $\boxed{\mathbb{Z} \ni 1}$

$\mathbb{Z} \ni 1$ $\quad$ $1 \in \mathbb{Z}$

"$\ni$" = "s.t."

$A \supseteq B$

$A \supseteq B$

__Lemma:__ Suppose $A \subseteq B$ and $C \subseteq D$.
Then $A \times C \subseteq B \times D$.

__Proof:__ Suppose $A \subseteq B$ and $C \subseteq D$. Let $(a,c) \in A \times C$.

Then $a \in A$ and $c \in C$.

(WTK: $a \in B$ and $c \in D$)

Since $A \subseteq B$, $a \in B$. Since $C \subseteq D$, $c \in D$.

Thus $(a,c) \in B \times D$. ∎

**Power Set:** Let $A$ be a set. Then we define the power set $P(A)$ to be

$$P(A) = \{B \text{ s.t. } B \subseteq A\}$$

**Examples:** $A = \{1,2\}$ $\quad P(A) = \{\{1\}, \{2\}, \phi, \{1,2\}\}$

$\{1\} \subseteq \{1,2\}$

$\{2\} \subseteq \{1,2\}$

$\emptyset \subseteq \{1,2\}$

$\{1,2\} \subseteq \{1,2\}$

$\{1\} \in P(\{1,2\})$

$1 \notin P(\{1,2\})$

Rule: $B \in P(A) \iff B \subseteq A$

$1 \notin P(\{1,2\})$ b/c $1 \not\subseteq \{1,2\}$

**Claim:** $\# P(A) = 2^{\# A}$

**Ex:** $A = \{7\}$ $\quad 1$

$P(A) = \{\phi, A\}$ $\quad 2^1$

**Ex:** $P(\phi) = \{B : B \subseteq \phi\}$

$= \{\phi\}$

$\# \phi = 0$

$\# \{\phi\} = 1 \qquad 1 = 2^0$

**Ex:** $\phi, A \in P(A)$ b/c $\phi \subseteq A$

$A \subseteq A$

$$P(\mathbb{Z}) = \{\phi, \mathbb{Z}, \mathbb{E}, 3\mathbb{Z}, 4\mathbb{Z}, \ldots d\mathbb{Z}, \ldots \\ \{1\}, \{2\}, \{3\}, \ldots \\ \{1,2\}, \{1,2,3\}, \ldots\}$$

$\mathbb{E} \in P(\mathbb{Z})$

$d\mathbb{Z} \in P(\mathbb{Z})$

$\mathbb{R} \notin P(\mathbb{Z})$ b/c $\mathbb{R} \not\subseteq \mathbb{Z}$